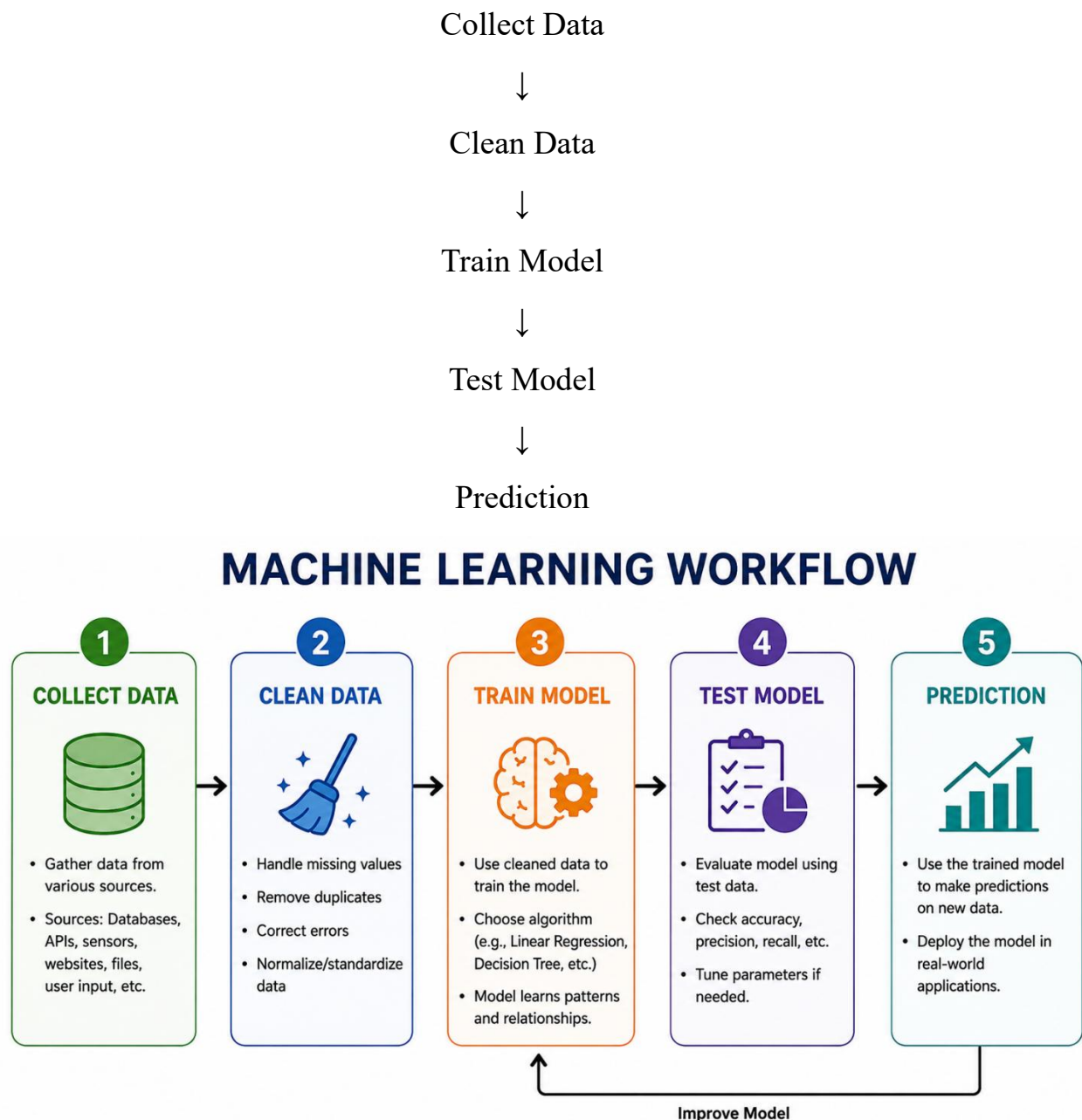


Task 6: ML Workflow

1. Draw ML workflow diagram
2. Explain each step
3. Give real-world example

Draw ML workflow diagram



This workflow explains how a Machine Learning model is created and used to make intelligent predictions.

Explain each step and Give real world examples

Step 1: Collect Data

Meaning

The first step in Machine Learning is collecting data from different sources.

Data is the foundation of every ML system.

Without data:

Machine Learning cannot work

Sources of Data

Data can come from:

- Websites
- Mobile apps
- Sensors
- Databases
- Surveys
- Cameras
- User activities

Examples of Data

Application	Data Collected
YouTube	Watch history
Hospital	Patient records
Shopping App	Purchase history
School	Student marks

Importance of Data Collection

Good quality data helps:

- Improve accuracy
- Train better models

- Make better predictions

Poor data can produce wrong results.

Step 2: Clean Data

Meaning

Collected data is usually not perfect.

It may contain:

- Missing values
- Duplicate records
- Incorrect data
- Empty rows
- Noise

Data cleaning removes these problems.

Example

Raw Data

Study Hours	Marks
2	40
4	Empty
4	Empty
6	90

Problems:

- Missing value
- Duplicate row

After Cleaning

Study Hours	Marks
2	40
4	70
6	90

Why Data Cleaning is Important

Clean data helps:

- Increase model performance
- Reduce errors
- Improve prediction quality

This is one of the most important steps in ML.

Step 3: Train Model

Meaning

The model studies:

- Patterns
- Relationships
- Trends in data

Example

Study Hours	Marks
2	40
4	70
6	90

The model learns:

More study hours → Higher marks

What Happens During Training?

The ML algorithm:

- Reads data repeatedly
- Identifies relationships
- Adjusts internal calculations
- Improves learning

This process creates a trained model.

Goal of Training

Teach the machine to learn patterns from data

Step 4: Test Model

Meaning

After training, the model is tested using new data.

The testing data is different from training data.

Purpose:

- Check model accuracy
- Measure performance
- Find mistakes

Example

New Input

Study Hours = 5

Expected Prediction

Marks $\approx$ 80

If predictions are close to actual values, the model works well.

Why Testing is Important

Testing helps:

- Detect errors
- Improve reliability
- Avoid wrong predictions

Without testing:

The model may not work properly in real situations

Step 5: Prediction

Meaning

Prediction is the final step of Machine Learning.

The trained model gives outputs for new inputs.

Example

Input

Study Hours = 7

Prediction

Predicted Marks = 95

The model uses learned patterns to predict results automatically.

Real-World Predictions

Application	Prediction
Email System	Spam or Not Spam
Shopping Apps	Product Recommendation
Banking	Fraud Detection
Weather Apps	Weather Forecast
Maps	Traffic Prediction

Real-World Example of ML Workflow

Spam Email Detection Workflow

Step 1: Collect Data

Thousands of emails are collected.

Step 2: Clean Data

Remove:

- Duplicate emails
- Empty messages
- Unwanted symbols

Step 3: Train Model

The ML model learns patterns of:

- Spam emails
- Normal emails

Step 4: Test Model

- The system checks prediction accuracy using new emails.

Step 5: Prediction

When a new email arrives:

- ML predicts Spam or Not Spam
- Spam emails move to spam folder automatically

Used in:

- Gmail
- Microsoft Outlook